Michael Borck School of Marketing and Management Curtin University Kent Street, Bentley, Perth, WA 6102, Australia michael.borck@curtin.edu.au

31 October 2026

Professor Gil Fellingham Editor-in-Chief *Journal of Quantitative Analysis in Sports*

Re: Submission of "Skill, Sequence, and Scoring: A Mathematical Comparison of Traditional and World Bowling Scoring Systems"

Dear Professor Fellingham,

I am pleased to submit the above manuscript for consideration as a Research Article in the *Journal of Quantitative Analysis in Sports*.

Element of novelty. To our knowledge, the paper provides the first formal mathematical comparison of traditional ten-pin bowling scoring and the current-frame scoring system promoted by World Bowling (now the IBF). The closest prior study (Abdel-Salam et al., 2017) measures whether a *different* alternative scoring system preserves player placements; it does not compare traditional with current-frame scoring, characterise sequence sensitivity, or separate raw-skill from sequencing information. Its contributions are: (1) an exact combinatorial enumeration of both score distributions — extending Cooper and Kennedy's (1990) classic result to a second scoring system and verifying against published tables to exact integer precision; (2) a formal proof that current-frame scoring is commutative (invariant to frame order), together with an exhaustive permutation analysis quantifying the sequence sensitivity of traditional scoring (up to a 75-point range across orderings of identical frames); and (3) a variance-decomposition reliability analysis, calibrated to published PBA data (McCarthy 2011; VanDerwerken and Kenter 2018), showing that the two systems are nearly identical in single-game reliability on raw skill, and that the exclusive competitive advantage of traditional scoring is its sensitivity to *sequencing* — the ability to string consecutive strikes — which current-frame scoring is structurally unable to record. The question is timely: current-frame scoring is being adopted in new international franchise competition and is central to the sport's Olympic inclusion bid, so governing bodies are making this design choice now, and this paper quantifies precisely what that choice keeps and discards.

I confirm that the manuscript is original, has not been published previously, and is not under consideration by any other journal. There are no competing interests to declare. All code, data, and analysis scripts are publicly available at https://github.com/michael-borck/skill-sequence-scoring, and an interactive companion site is available at https://michael-borck.github.io/skill-sequence-scoring.

Manuscript details: approximately 6,700 words of main text; 10 tables; 13 figures; no supplementary material. AI-assisted tools were used in code development and drafting and are disclosed in the Acknowledgements; all mathematical results, research design, and editorial decisions are my own.

Thank you for your consideration. I look forward to the reviewers' comments.

Sincerely,

Michael Borck School of Marketing and Management, Curtin University